

Flange load cell PD 21 / PD 22

- Position-independent installation due to various mounting options such as flange bearings, pedestal bearings, inner or outer fastening and central fastening
- High reliability due to overload protection up to 20 times the nominal measuring force
- Different shaft diameters from 12 to 65 mm and nominal measuring forces from 0.05 to 10 kN ensure a high degree of flexibility
- With a horizontal measuring direction, the weight of the roller does not affect the measured result
- Good temperature behavior and a high degree of linearity of the measuring elements due to strain gauge application on a flat surface
- High permissible operating speed of the measuring roller due to high web load cell spring constant
- Best surface protection due to chemical nickel plating



Flange load cell PD 21 with accessory kit 7.1 (C-clamp)

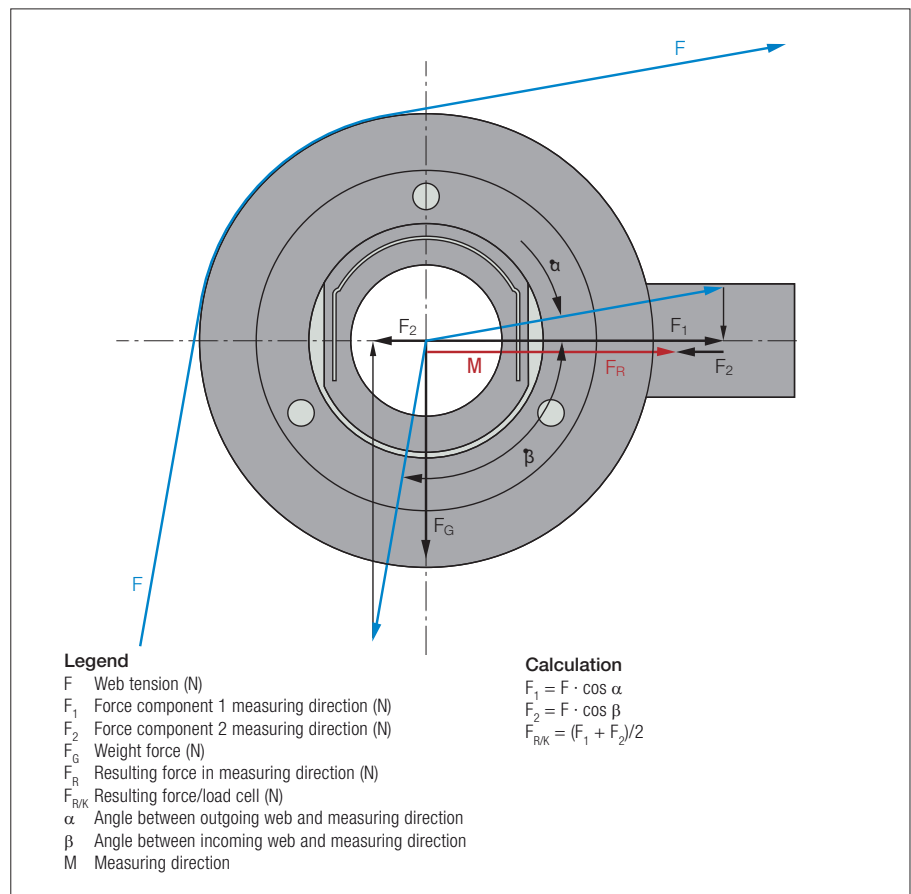


Function

The load cell comprises a stable outer ring with flange cover and centering collar for precision assembly. The inner ring in the form of a double bending beam ensures centered mounting of the ball bearing. The radial forces created by the web unbalance the strain gauges linked together to form a measuring bridge on the inner ring. This leads to an analog output signal proportional to the web tension.

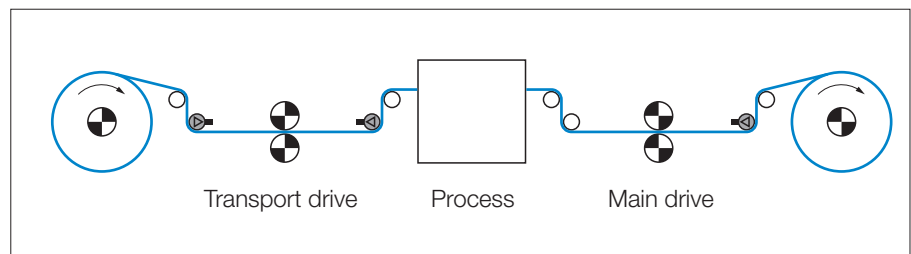
Area of use

Flange load cells are used in practically all processing plants where web-type materials are processed or finished. In front of processing stations, in particular, it is of fundamental importance that the web is transported with a continuous web tension.



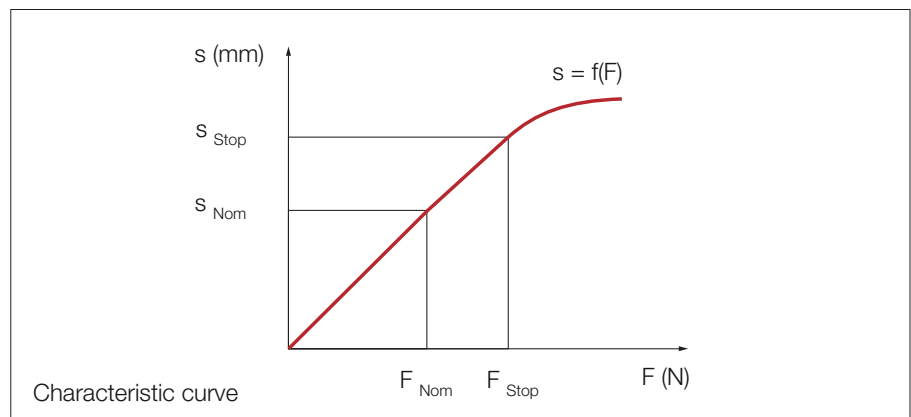
Application

With a 90° horizontal-vertical wrapping angle on the measuring roller and a horizontal measuring direction, optimum web tension detection is assured. Only detection of the bearing forces on both sides can prevent incorrect measurements caused by the web moving sideways and asymmetrical web tension distribution. Load cells incorporated into a control loop should be mounted as near to the actuator as possible.



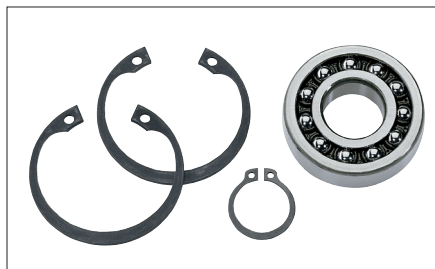
Calibration

The tensile force-path characteristic curve forms a straight line to the mechanical stop. All load cells are calibrated to the nominal measuring force. Between the nominal measuring force and mechanical stop, a safety factor of 50 to 100 % is taken into account to compensate for asymmetrical web tension distribution.

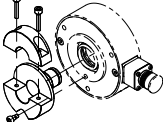


Accessory kits

Accessory kits with self-aligning ball bearing, securing ring and compensating sleeves for the different assembly variants



Selection table for accessory kits

Roller version	Shaft/axle version	For load cell	Floating bearing side accessory kit	Fixed bearing side accessory kit
	Stepped axle	PD 21	1	3
	Continuous axle	PD 21	2	4
	Stepped shaft	PD 21	1	1
	Continuous shaft	PD 21	2	2
	Stepped shaft	PD 22	5	5
	Continuous shaft	PD 22	6	6
	C-clamp	PD 21	7.1	7.1

Mounting bracket

for precision mounting of flange load cells on machine frames

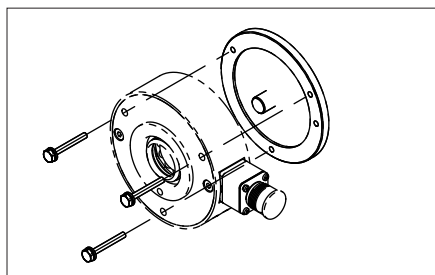


Selection table for mounting brackets

Material number	Shaft diameter (mm)												
	12	15	17	20	25	30	35	40	45	50	55	60	65
042594	X	X	X										
042595				X	X								
042596						X	X						
042597								X	X	X			
042598											X	X	X

Mounting flange

for mounting flange load cells with one central screw only



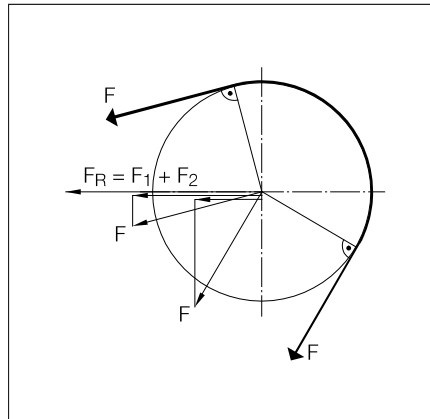
Selection table for mounting flanges

Material number	Shaft diameter (mm)												
	12	15	17	20	25	30	35	40	45	50	55	60	65
514958					X								
515033							X						

Selection table for cables

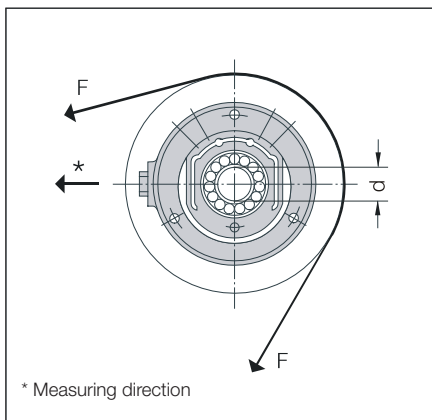
Material number	Designation	Length (m)
504024	PDV0200 – 57	5
511666	PDV0200 – 57	10
831933	PDV0200 – 57	15
831934	PDV0200 – 57	20
831935	PDV0200 – 57	25
831936	PDV0200 – 57	30

Calculation



You can calculate the load cell values using our free calculator.

Software download
 "ELTENS-Calculator":



Technical data PD 21 / PD 22

Accuracy class	0.5
Nominal characteristic value (sensitivity)	1 mV/V
Combined error (hysteresis/non-linearity)	< 0.5 %
Characteristic value tolerance	0.2 %
Measuring principle	full bridge strain gauge
Nominal resistance of strain gauge bridge	700 Ohm
Bridge supply voltage	
- Nominal value	10 V
- Max. permissible value	14 V
Mechanical stop	1.8 to 2.4 x F _N depending on the type
Operating load	1.8 to 2.4 x F _N
Limit load	20 x F _N
Nominal measuring deflection	0.1 to 0.2 mm depending on the type
Nominal temperature range	-10 to +60 °C
Normal temperature range	-10 to +90 °C
Temperature coefficient	
- of the characteristic value	±0.3 % / 10 K
- of the zero signal	±0.3 % / 10 K
Protection class	IP 50
Max. permissible axial lateral force	1 x F _N
Weight	2.3 kg (d = 17 mm) 3.6 kg (d = 25 mm) 8.5 kg (d = 35 mm)

Subject to technical change without notice

Selection table

Type Bore on one side	Type Bore on both sides	d (mm)	Nominal measuring force (kN)					
PD 2112	PD 2212	12	0.05	0.1	0.2	0.5	1	
PD 2115	PD 2215	15	0.05	0.1	0.2	0.5	1	
PD 2117	PD 2217	17	0.05	0.1*	0.2	0.5*	1	
PD 2120	PD 2220	20		0.15	0.3	0.75	1.5	
PD 2125	PD 2225	25		0.15*	0.3	0.75*	1.5	
PD 2130	PD 2230	30		0.3	0.6	1.5	3	
PD 2135	PD 2235	35		0.3*	0.6	1.5*	3	
PD 2140	PD 2240	40		0.6	1.2	3	6	
PD 2145	PD 2245	45		0.6	1.2	3	6	
PD 2150	PD 2250	50		0.6*	1.2	3*	6	
PD 2155	PD 2255	55		1	2	5	10	
PD 2160	PD 2260	60		1	2	5	10	
PD 2165	PD 2265	65		1	2	5	10	

* Preferred sizes